

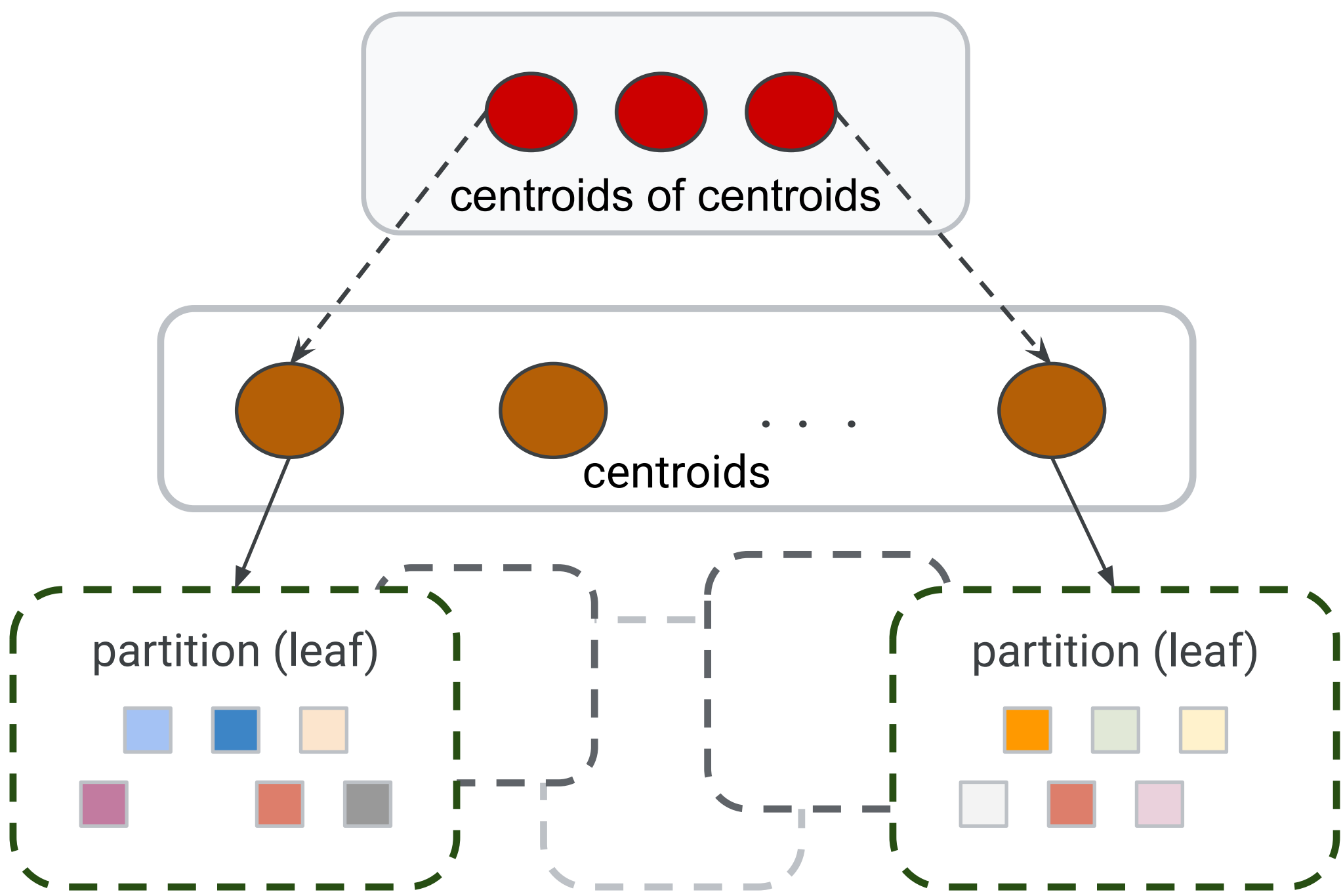


Bringing Vector Search to OLTP

Vector Search in Cloud SQL for MySQL
<https://docs.cloud.google.com/sql/docs/mysql/vector-search>



ScaNN (Scalable Nearest Neighbors)
state-of-the-art ANN search engine powering many planet scale Google products



{partition, vector_id} primary key
ensure the data are grouped by partition

{vector_id} unique key locate the exact row in the sub_table that corresponds to the base table row.

partition	vector_id	vector_data
-1	null	centroids
0	0	...
0	1	...
...		
k	m	...
k	n	...

persistence

Overall Cost:

- Find partitions that are *close enough*
- Scan the selected partitions to find the nearest neighbors

Scaling:

- More vectors → More partitions
- Multiple Levels (favor smaller partitions)

query ↓
centroids part of the index resides in memory

↓
a fraction of partition part of the index that are loaded dynamically at query time

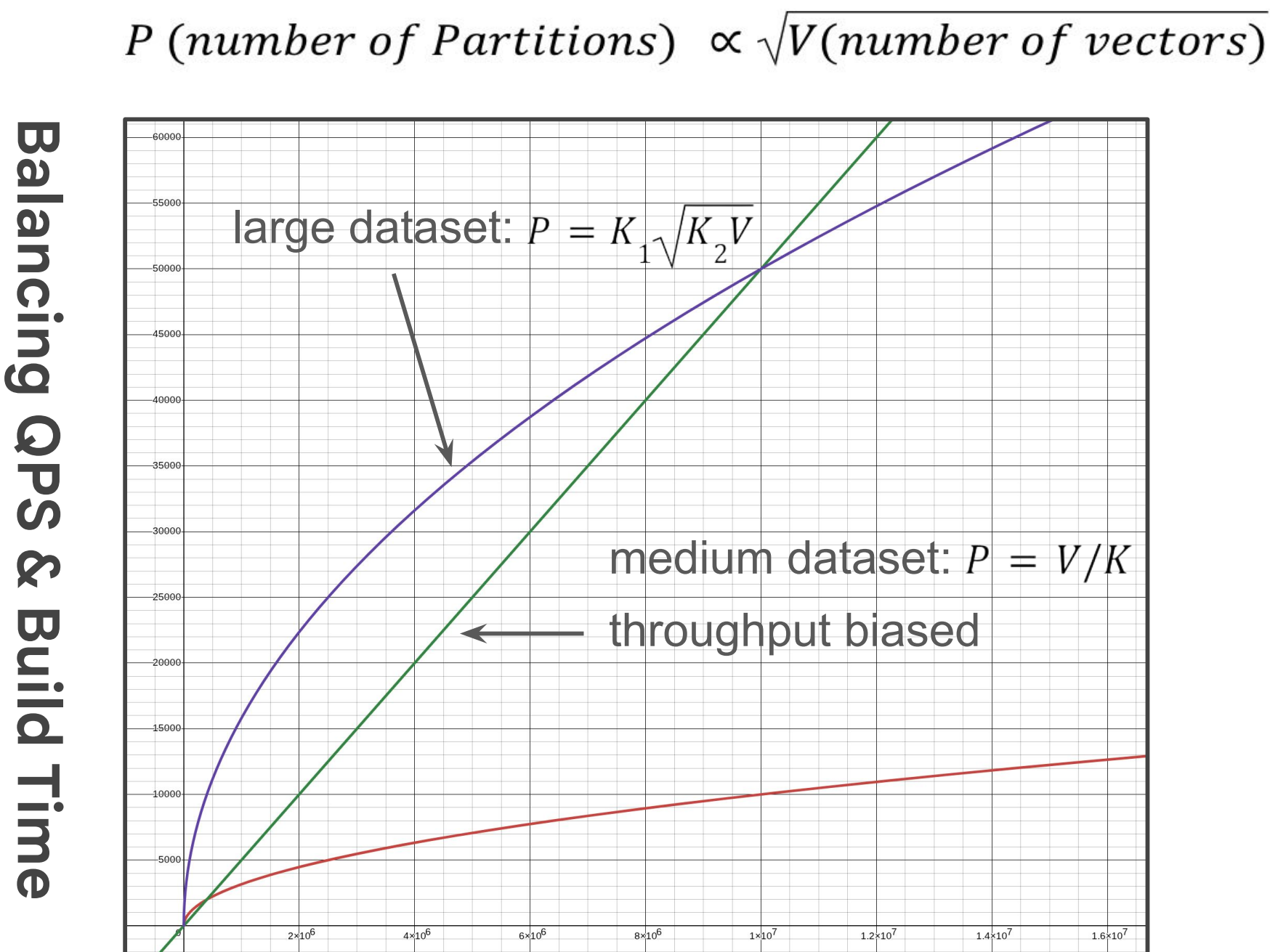
in-memory structure

Index → K-Means partitioning

Build = train (compute centroids) + insert (add data points - vectors)

- Each partition (leaf) has a certain number of vectors
- Vectors can be quantized (compressed) which makes ANN query much faster
- Supports common distance measures (cosine, euclidean, dot_product)

Bridging ANN and SQL	
Data Size Limit	N/A
DML	Yes
Crash Safety	Yes
MVCC	Fully transactional
Generate Embeddings	Yes
CURD Syntax	SQL compatible
Optimizer awareness	Yes
Hybrid Search: ANN & SQL	Iterative Search



Native SQL Search

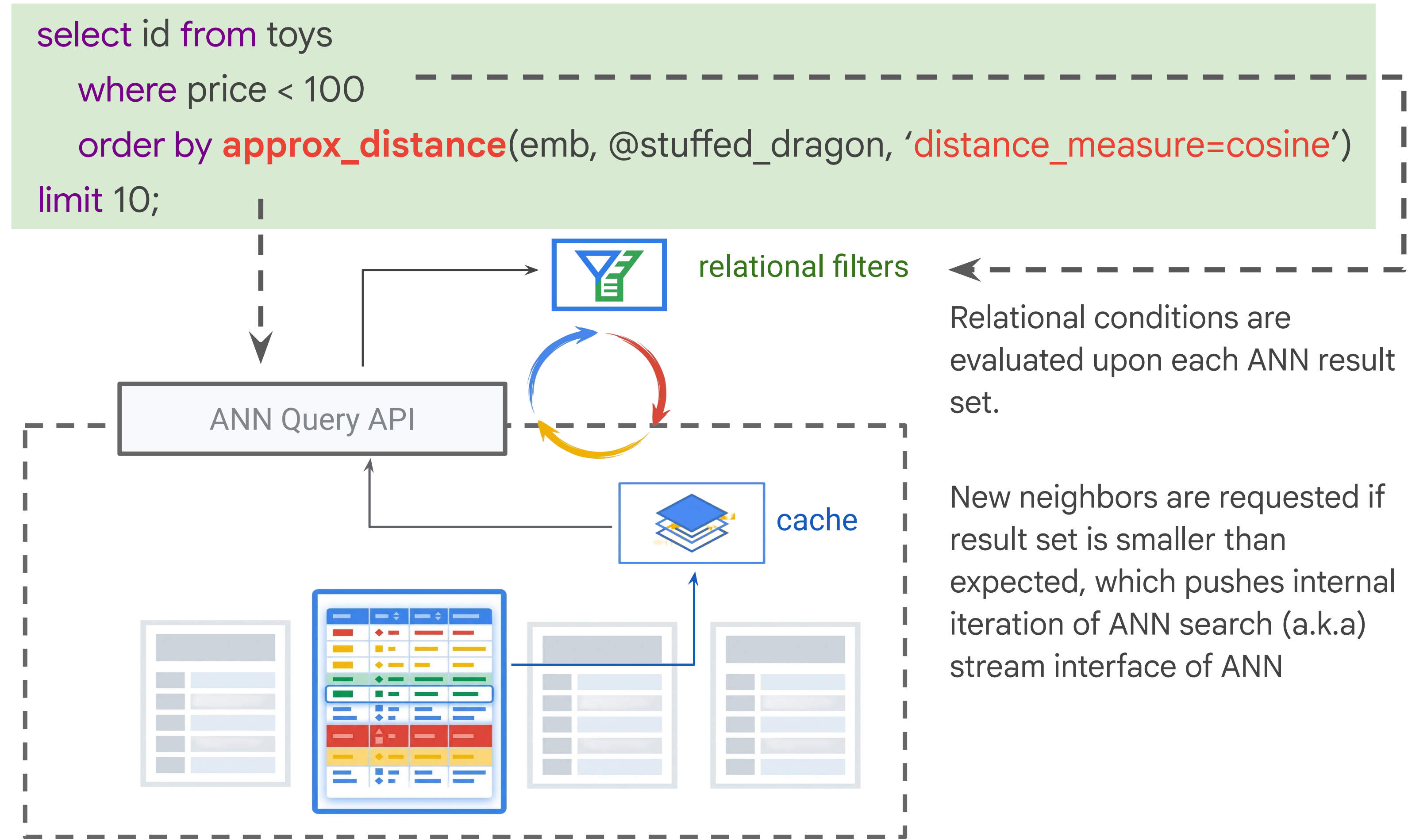
Managing Vector Data

Tables	create table toys (, ..., specs varchar(3000), emb vector(3072) using varbinary);	
Data	update toys set emb = mysql.ml_embedding('gemini-embedding-001', specs);	insert into toys values(1, "stuffed dragon", string_to_vector('[1,2,3,...]'));
Indexes	create vector index toy_specs_idx on toys (emb) distance_measure=cosine;	

Semantic Search

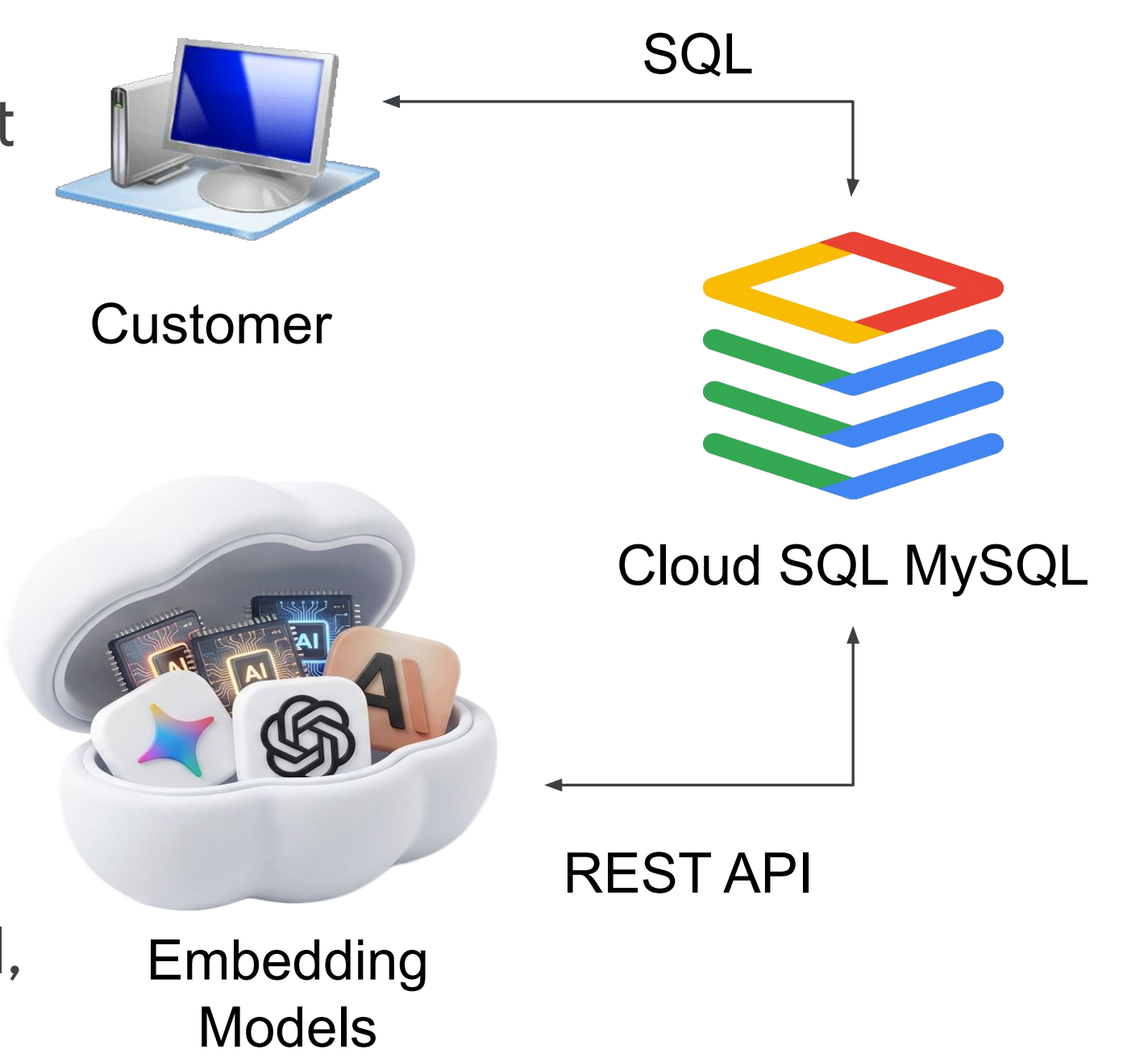
ANN	select id, approx_distance(emb, @stuffed_dragon, 'distance_measure=cosine') similarity from toys order by similarity limit 10;
KNN	select id from toys order by cosine_distance(emb, @stuffed_dragon, 'distance_measure=cosine') limit 10;

Hybrid Search



Model Endpoint Management

- Customer**
- Customers want to compute embeddings directly from SQL, without external pipelines.
- Model Endpoint Management**
- Allows forwarding requests to models from different providers to generate embeddings
 - Customers can get online predictions from models hosted in Google (Vertex AI) and other providers such as OpenAI, Anthropic, etc.



Fully Managed. Gen AI Ready.

Cloud SQL for MySQL handles the heavy lifting of database administration so you can focus on innovation. Experience the reliability of a trusted platform combined with the power of built-in vector search. Transform your operational data into a Gen AI powerhouse — generating embeddings and running similarity search using standard SQL, with no separate vector database required.



Cloud SQL for MySQL